

Fall 2020 CS215 Quiz 1

Q1 The prob of type $F_1, F_1 = p^2$
 " " " " $F_1, F_2 = 2p(1-p)$
 " " " " $F_2, F_2 = (1-p)^2$

The likelihood function of $n_1, n_2, n_3 = n - n_1 - n_2$ given p is

$$\frac{n!}{n_1! n_2! (n - n_1 - n_2)!} (p^2)^{n_1} (2p(1-p))^{n_2} ((1-p)^2)^{n - n_1 - n_2}$$

$$= \frac{n!}{n_1! n_2! (n - n_1 - n_2)!} p^{2n_1} (2p(1-p))^{n_2} (1-p)^{2(n - n_1 - n_2)}$$

This is a multinomial distribution with parameters $p_1 = p^2, p_2 = 2p(1-p)$ and $p_3 = (1-p)^2$

Note that $p_1 + p_2 + p_3 = p^2 + 2p - 2p^2 + 1 + p^2 - 2p = 1$

$$\begin{aligned} \log \text{likelihood} &= +2n_1 \log p + n_2 \log (2p(1-p)) \\ &+ \text{constants} + 2(n - n_1 - n_2) \log (1-p) \\ &= \log p (2n_1 + n_2) + (n_2 + 2n_3) \log (1-p) \end{aligned}$$

$$\frac{\partial LL}{\partial p} = \frac{2n_1 + n_2}{p} - \frac{n_2 + 2n_3}{1-p} = 0$$

$$\therefore (2n_1 + n_2)(1-p) = (n_2 + 2n_3)p$$

$$\therefore 2n_1 + n_2 = p(2n_1 + 2n_2 + 2n_3)$$

$$\therefore p = \frac{2n_1 + n_2}{2n}$$



$$Q_2 \quad RHS = \sum_{i=0}^{\infty} P(Y > i)$$

$$= P(Y > 0) + P(Y > 1) + P(Y > 2) + \dots$$

$$= P(Y=1) + P(Y=2) + P(Y=3) + P(Y=4) + \dots$$

$$+ P(Y=2) + P(Y=3) + P(Y=4) + \dots$$

$$+ P(Y=3) + P(Y=4) + \dots$$

$$= 1P(Y=1) + 2P(Y=2) + 3P(Y=3) + \dots$$

$$= \sum_{i=0}^{\infty} i P(Y=i) = E(Y).$$

So statement (c) is true.



$$Q3 \quad P(X_1 X_2 \leq x) \quad (X_1, X_2 \text{ are indep. r.v.s})$$

$$= P(X_1 X_2 \leq x, X_1 \geq 0) + P(X_1 X_2 \leq x, X_1 < 0)$$

$$= P\left(X_2 \leq \frac{x}{X_1}, X_1 \geq 0\right) + P\left(X_2 \geq \frac{x}{X_1}, X_1 < 0\right)$$

$$= \int_0^{\infty} f_{X_1}(x_1) \int_{-\infty}^{x/x_1} f_{X_2}(x_2) dx_2 dx_1$$

$$+ \int_{-\infty}^0 f_{X_1}(x_1) \int_{x/x_1}^{\infty} f_{X_2}(x_2) dx_2 dx_1$$

$$f_{X_1 X_2}(x) \stackrel{*}{=} \int_0^{\infty} f_{X_1}(x_1) \frac{d}{dx} \left[F_{X_2}\left(\frac{x}{x_1}\right) - F_{X_2}(-\infty) \right] dx_1$$

$$+ \int_{-\infty}^0 f_{X_1}(x_1) \frac{d}{dx} \left[F_{X_2}(\infty) - F_{X_2}\left(\frac{x}{x_1}\right) \right] dx_1$$

$$= \int_0^{\infty} f_{X_1}(x_1) f_{X_2}\left(\frac{x}{x_1}\right) \frac{1}{x_1} dx_1$$

$$+ \int_{-\infty}^0 f_{X_1}(x_1) f_{X_2}\left(\frac{x}{x_1}\right) \frac{(-1)}{x_1} dx_1$$


$$= \int_{-\infty}^{\infty} f_{X_1}(x_1) f_{X_2}\left(\frac{x}{x_1}\right) \frac{1}{|x_1|} dx$$

$$Q4 \quad H(X) = - \sum_{i=1}^K p_i \log p_i$$

$$= - \sum_{i=1}^{K-1} p_i \log p_i - \left[1 - \sum_{i=1}^{K-1} p_i \right] \log \left[1 - \sum_{i=1}^{K-1} p_i \right]$$

$$\frac{\partial H(X)}{\partial p_j} = -[1 + \log p_j] \text{ for } 1 \leq j \leq K-1$$

$$- \left[-1 + \log \left(1 - \sum_{i=1}^{K-1} p_i \right) (-1) \right]$$

$$= -\log p_j + \log \left[1 - \sum_{i=1}^{K-1} p_i \right]$$

$$\therefore p_j = 1 - \sum_{i=1}^{K-1} p_i = p_K$$

This holds true for all $1 \leq j \leq K-1$. This implies that $\forall j, 1 \leq j \leq K$, the p_j values are equal. As the $\{p_j\}_{j=1}^K$

add up to 1, we have

$$\forall j, 1 \leq j \leq K, p_j = 1/K.$$

Thus a uniform PMF is the one with maximum entropy. To verify it is the maximum, we do a double derivative test

$$\frac{\partial^2 H}{\partial p_j^2} = \frac{-1}{p_j} + \frac{(-1)}{1 - \sum_{i=1}^{K-1} p_i} < 0$$

Q5 If $p = 0$

$$f_Z(z; p) = \begin{cases} 1 & 0 < z < 1 \\ 0 & \text{otherwise} \end{cases}$$

If $p = 1$,

$$f_Z(z; p) = \begin{cases} 1/2\sqrt{z} & 0 < z < 1 \\ 0 & \text{otherwise} \end{cases}$$

If $p = 0$, the joint likelihood of $\{Z_i\}_{i=1}^n$ is
$$\prod_{i=1}^n 1(0 < Z_i < 1) = 1$$
 as

all Z_i have support betn. 0 and 1.

If $p = 1$, the joint likelihood is

$$\prod_{i=1}^n \frac{1}{2\sqrt{Z_i}} = 2^{-n} \prod_{i=1}^n \frac{1}{\sqrt{Z_i}}$$

$\therefore$ If $2^{-n} \prod_{i=1}^n Z_i^{-1/2} > 1$, then
the MLE $\hat{p} = 1$, otherwise
 $\hat{p} = 0$

Note that Z_i is the value of Z_i for $\forall i, 1 \leq i \leq n$.